NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

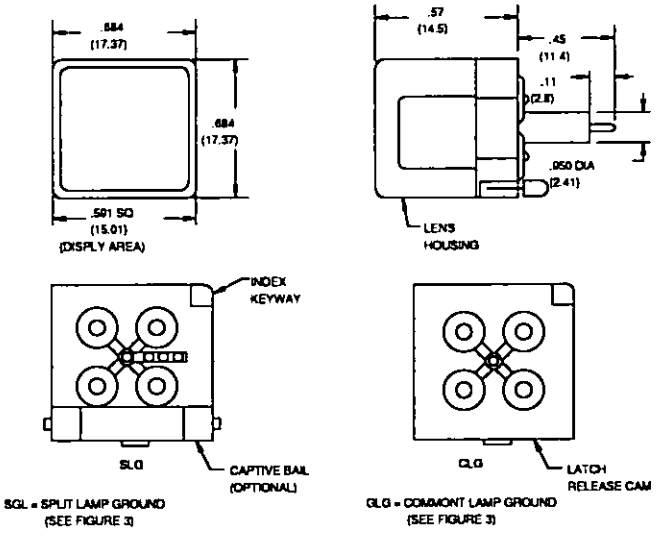
MIL-PRF-22885/104A(EC)
10 OCTOBER 1989
SUPERSEDING
MIL-S-22885/104EC
15 October 1985

PERFORMANCE SPECIFICATION SHEET

SWITCH, PUSHBUTTON, ILLUMINATED, 4 LAMP, SUNLIGHT READABLE
(FOR ILLUMINATED PUSHBUTTON SWITCH)

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-22885.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.51 mm) for two place decimals.
5. Index keyway prevents incorrect insertion into MIL-S-22885/103 switch housing.
6. Pushbuttons are available with captive pushbutton feature, i.e., pushbutton remains captive to MIL-S-22885/103 switch housing during relamping (see figure 2).

FIGURE 1. Pushbutton dimensions and configuration.

Ⓐ denotes changes

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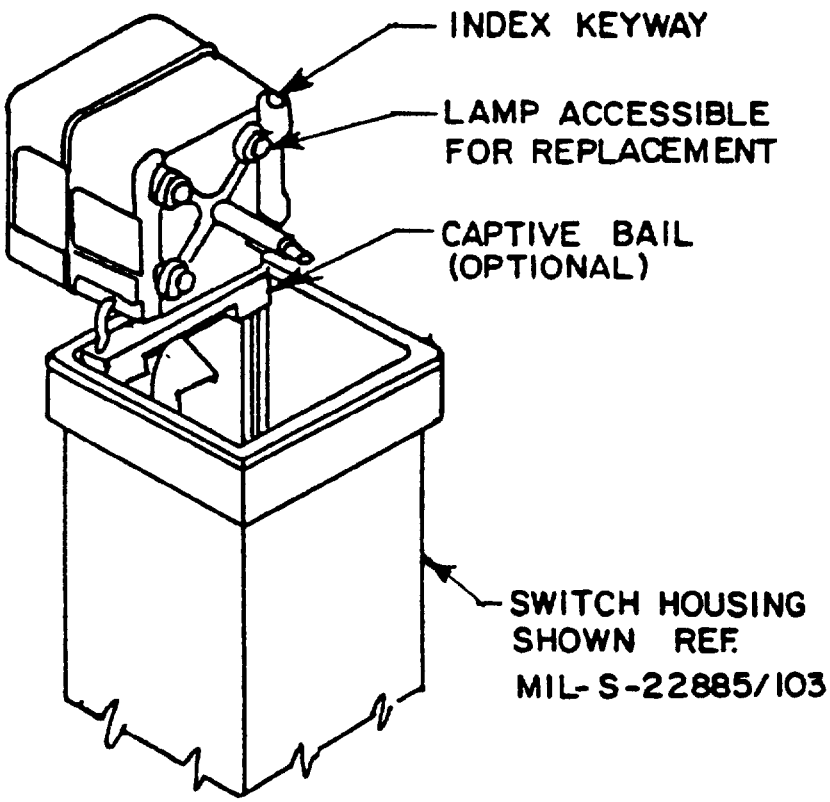
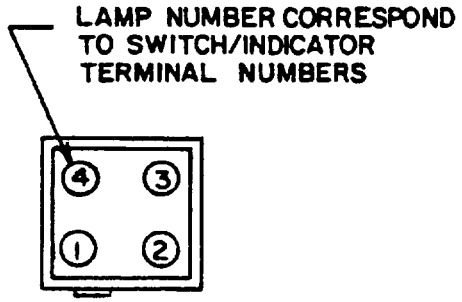
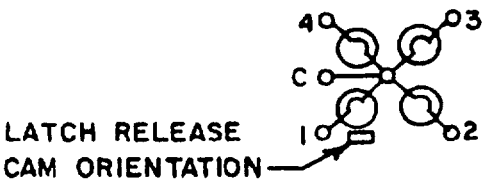


FIGURE 2. Captive pushbutton feature.

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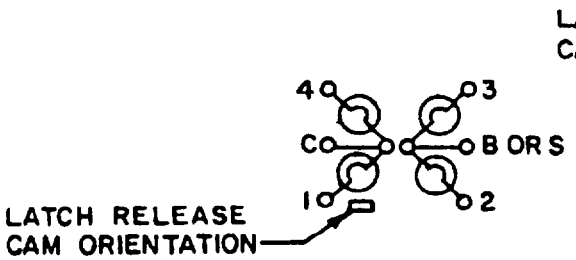


LENS, FRONT VIEW

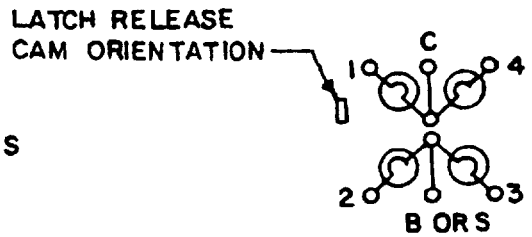


DISPLAY STYLE: ALL
(SEE FIG 4)

COMMON LAMP GROUND
CLG



DISPLAY STYLE: F, G, & H
(SEE FIG 4)



DISPLAY STYLE: A, B, C, D & E
(SEE FIG. 4)

SPLIT LAMP GROUND
SLG

FIGURE 3. Lamp circuit schematics.

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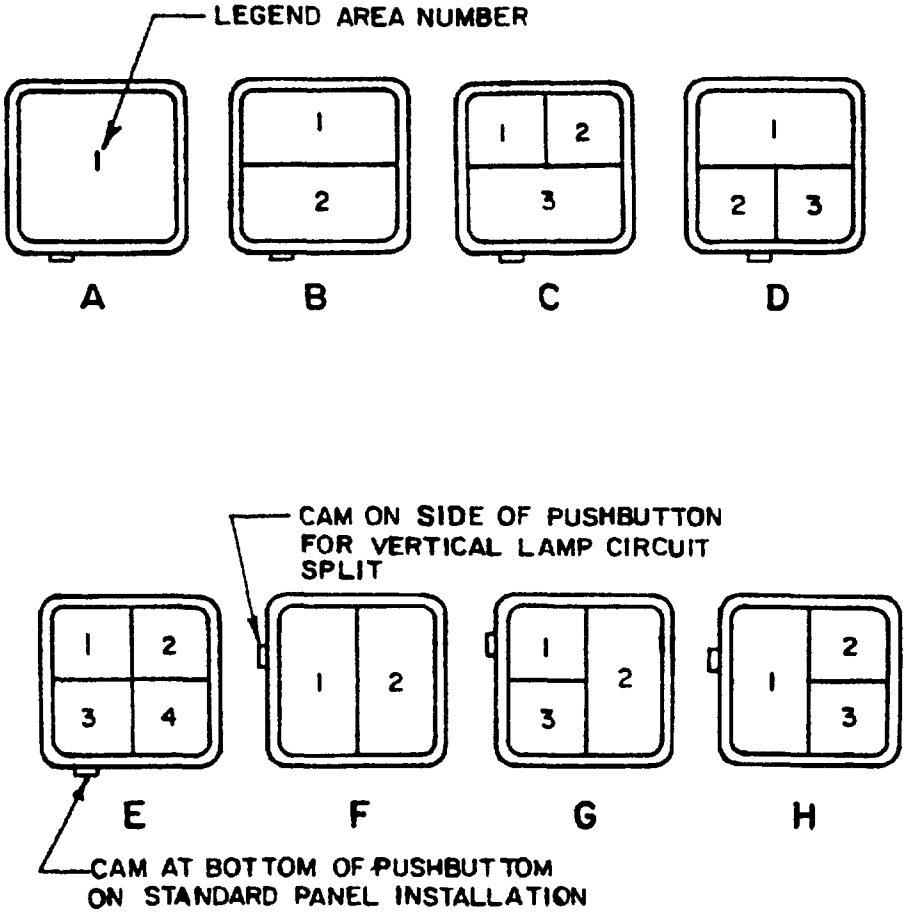


FIGURE 4. Lens configuration and legend areas.

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REQUIREMENTS:

Design and construction: See figures 1 through 4.

Intended use: These pushbuttons are used in conjunction with switch or indicator housings in accordance with MIL-S-22885/103 or multistation housings in accordance with MIL-S-24317/12. They accept four T-1 midget flange base lamps with .15 mean spherical candle power. They are intended for use in illuminated switch/indicators under high ambient light conditions. Displays with contrast ratios of .6 or higher in lighted condition and .1 or lower in unlighted condition (see table I) are considered readable under direct sunlight conditions.

Classification:

Enclosure design: 1 (Unsealed).

Temperature characteristic: 1 (-55°C to +85°C).

Vibration grade: 3 (10 - 2,000 Hz, 15 g peak).

Color: See table I.

Display type:

S (Sunlight readable).

H (Hidden legend).

Material:

Pushbutton housing: Alluminum alloy, black anodize finish.

Display screen and lamp module: Thermoplastic.

Weight: .158 ounce (4.5 grams) ±10 percent including lamps.

Shock: Method I.

Dielectric withstanding voltage:

1,000 V rms sea level.

450 V rms 70,000 feet.

EMI/RFI shielding: Positive grounding and shielding efficiency test applicable when specified, see table II. See MIL-S-22885/103 for minimum shielding attenuation.

Illuminated color: See table II.

Luminance: See table II (values are for reference only).

Ⓐ Sunlight readability: Applicable, except units are tested for $\phi_1 = 45°$ and $\phi_2 = 0°$ only. (This procedure does not test readability in direct reflected specular sunlight.) See table II for required contrast ratio.

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TABLE II. Dash numbers and characteristics. 1/.

Dash no.	Lens configuration (see fig. 4)	EMI shield	Captive bail (see fig. 2)	Circuit (see fig. 3)
T001	A	Yes	Yes	CLG
T002	"	"	Yes	SLG
T003	"	"	No	CLG
T004	"	"	No	SLG
T005	"	No	Yes	CLG
T006	"	"	Yes	SLG
T007	"	"	No	CLG
T008	"	"	No	SLG
T009	B	Yes	Yes	CLG
T010	"	"	Yes	SLG
T011	"	"	No	CLG
T012	"	"	No	SLG
T013	"	No	Yes	CLG
T014	"	"	Yes	SLG
T015	"	"	No	CLG
T016	"	"	No	SLG
T017	C	Yes	Yes	CLG
T018	"	"	Yes	SLG
T019	"	"	No	CLG
T020	"	"	No	SLG
T021	"	No	Yes	CLG
T022	"	"	Yes	SLG
T023	"	"	No	CLG
T024	"	"	No	SLG
T025	D	Yes	Yes	CLG
T026	"	"	Yes	SLG
T027	"	"	No	CLG
T028	"	"	No	SLG
T029	"	No	Yes	CLG
T030	"	"	Yes	SLG
T031	"	"	No	CLG
T032	"	"	No	SLG
T033	E	Yes	Yes	CLG
T034	"	"	Yes	SLG
T035	"	"	No	CLG
T036	"	"	No	SLG
T037	"	No	Yes	CLG
T038	"	"	Yes	SLG
T039	"	"	No	CLG
T040	"	"	No	SLG
T041	F	Yes	Yes	CLG
T042	"	"	Yes	SLG
T043	"	"	No	CLG
T044	"	"	No	SLG
T045	"	No	Yes	CLG
T046	"	"	Yes	SLG
T047	"	"	No	CLG
T048	"	"	No	SLG
T049	G	Yes	Yes	CLG
T050	"	"	Yes	SLG
T051	"	"	No	CLG
T052	"	"	No	SLG
T053	"	No	Yes	CLG
T054	"	"	Yes	SLG
T055	"	"	No	CLG
T056	"	"	No	SLG
T057	H	Yes	Yes	CLG
T058	"	"	Yes	SLG
T059	"	"	No	CLG
T060	"	"	No	SLG
T061	"	No	Yes	CLG
T062	"	"	Yes	SLG
T063	"	"	No	CLG
T064	"	"	No	SLG

1/ Military part numbers do not define legend area colors or legend information. The sequence in which the color and legend information is specified is in accordance with the area numbers shown on figure 4. Additional documentation is required to acquire these pushbuttons.

CONCLUDING MATERIAL

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-N669)

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